Private AI Demo App Plan

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Summary

Build a small native Mac app, **Lone Star ITS Private AI (Demo)**, that chats with an open-weight AI model running entirely on the MacBook Air, with no internet and no ChatGPT or Claude behind it. It is the laptop-scale version of the Office AI tier: same idea, same kind of model, smaller box.

- **Stack:** a Tauri desktop app (web-style UI in a small native shell) that starts a bundled llama.cpp engine on the laptop and talks to it over 127.0.0.1 only.
- **Model:** chosen by RAM. On a 16GB Air, an 8B model such as Qwen3 8B; on 24GB or more, gpt-oss-20b, the smaller sibling of the 120B model the Office AI box runs.
- **First milestone:** a branded chat window that streams answers with Wi-Fi turned off. That alone carries the core pitch: "your data never leaves this machine."
- **Where it lives:** a new private repo, not the public website repo.

Plan only; nothing is built yet. Open questions for Zeke are at the end.

Goals and the demo story

The app exists to make one claim believable in under five minutes: a useful AI assistant can run on hardware the client owns, with nothing sent to the cloud.

Goals:

- Works with the network off, every time, so the demo never depends on venue Wi-Fi.
- Looks like a Lone Star ITS product, not a hobby tool, using the website's colors and logo.
- Does real office work a law or accounting firm recognizes: summarize, draft, rewrite, answer questions about a document.
- Stays simple enough to build and maintain alone.

A five-minute client demo:

1. Open the app. The header shows a green **Offline · Local only** badge and the model name.
2. Turn Wi-Fi off in front of the client.

3. Paste a sample engagement letter and ask for a plain-English summary and the key dates.
4. Ask it to draft a reply email to the client in a professional tone.
5. (Phase 2) Drop in a PDF and ask questions about it; answers cite the page they came from.
6. Close with: "This is a laptop. Your office box is a Mac Studio running a much larger model, for your whole team."

Use sample documents only in demos, never a real client's files.

Features

Version 1 is deliberately small: one chat window that works offline. Everything else waits until that is solid.

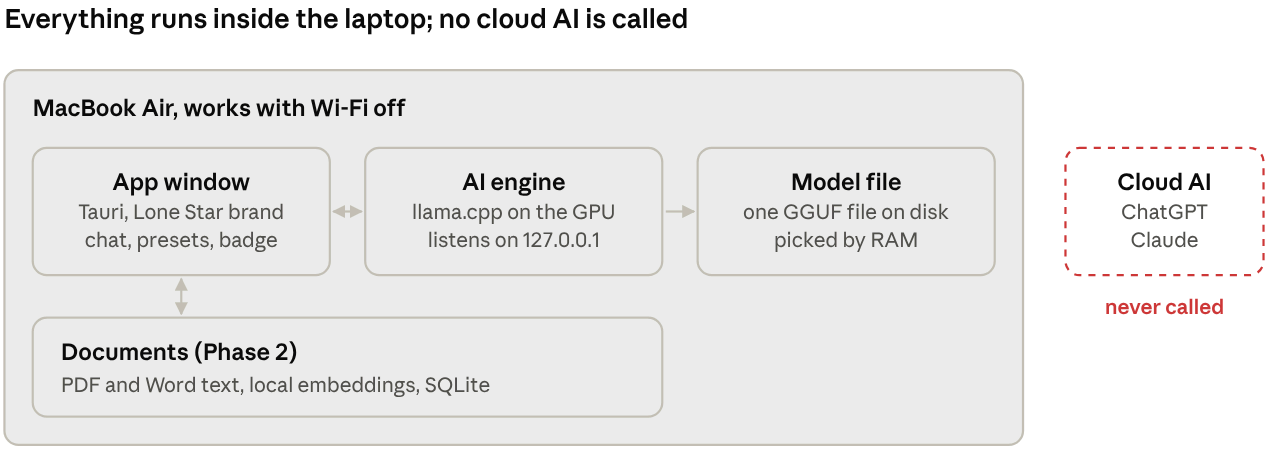
Feature	Phase	Why it matters in a demo
Chat with streamed answers	1	The basic "it just works" moment
Offline · Local only badge (checks the engine is on 127.0.0.1)	1	Makes the privacy claim visible
Lone Star ITS branding, light and dark	1	Looks like a product
Preset prompts: Summarize, Draft email, Rewrite plainly, Extract dates	1	One click to a good result, no prompt typing in front of a client
New chat / clear chat; nothing saved to disk by default	1	Nothing left behind on the laptop
Model name, size and speed (tokens per second) in a status bar	1	Answers "how fast is it" honestly
Drag in a PDF, Word or text file and ask about it, with page citations	2	The strongest pitch for law and accounting firms
Saved chat history, stored locally and optionally encrypted	2	Shows how data stays on the client's machine
Model picker (small and fast vs larger and smarter)	3	Explains the tiers: bigger box, bigger model

Feature	Phase	Why it matters in a demo
Signed, notarized .app and a one-click installer	3	Opens without macOS security warnings

Out of scope: accounts, cloud sync, web search, voice, and any call to an outside AI service.

Technical approach

The app is three local pieces: a window, an engine, and a model file. Nothing in the chain opens a connection off the laptop.



demo app architecture · 3 local parts, no cloud

The window talks only to the engine on 127.0.0.1, the engine reads the model from disk, and nothing crosses to the internet.

- **App shell: Tauri 2.** A small native macOS app (about 10 MB before the model) whose screens are written in HTML, CSS and TypeScript, so it can reuse the website's look. A thin Rust layer starts and stops the engine.
- **Engine: llama.cpp `llama-server`, bundled inside the app.** It runs on Apple Silicon's GPU through Metal and listens only on 127.0.0.1. It speaks the same OpenAI-style chat API as Ollama and vLLM, so the UI code carries over to the Business tier's vLLM server later.
- **Model: one GGUF file** in the app's Application Support folder, downloaded once during setup and then used offline.

- **Phase 2 documents:** text pulled from PDF and Word files locally, split into chunks, embedded with a small local embedding model, and stored in SQLite on disk.

Why not the alternatives:

- **Ollama plus Open WebUI** needs no code and is what the Office AI box runs, but it shows someone else's product. It is still a good backup demo.
- **SwiftUI plus Apple's MLX** is the fastest option on a Mac, but it locks the app to Apple and is harder to keep up alone.
- **Electron** works the same way as Tauri but ships a 150 MB+ app and uses more of the RAM the model needs.

Model choice depends on RAM

macOS lets the GPU use roughly two thirds to three quarters of memory, and the OS and app need the rest. The model file must fit comfortably inside that share, with room for the conversation.

MacBook Air RAM	Recommended model	Approx. file size	Notes
16 GB	Qwen3 8B, 4-bit (Q4_K_M)	about 5 GB	Default assumption. Quick, good at summaries and drafting; keep other apps closed
16 GB (alt)	Gemma 3 12B, 4-bit	about 7–8 GB	Better writing, slower, tighter on memory
24 GB	gpt-oss-20b	about 12–13 GB	Same family as the 120B model on the Office AI box: the best story
32 GB	gpt-oss-20b with longer context, or Qwen3 30B-A3B, 4-bit	about 13–18 GB	Room for long documents in Phase 2

Sizes are approximate and from memory; check them, and whether newer models have come out, before downloading. Qwen3 and gpt-oss are Apache 2.0, the simplest license for commercial demos. The Air has no fan, so long sessions slow down as it warms; short demos are fine.

Where the code lives

Recommendation: a new **private** repo, `lone-star-private-ai-app`, under ElSpaniard97. The custom-ai-models repo is public and GitHub Pages serves its root as the marketing site, so app code there would be visible to anyone and would clutter the site. The website can link to a download page later.

```
lone-star-private-ai-app/
  src/                 UI: chat screen, presets, status bar (TypeScript)
  src/styles/          brand colors and fonts from the website's style.css
  src-tauri/           Rust: start/stop the engine, find the model file
  src-tauri/binaries/  bundled llama-server for Apple Silicon
  scripts/             download-model.sh, build-mac.sh
  demo/                sample documents and the demo script
  README.md
```

How building works: code is written and pushed from here, but the Mac app has to be compiled and tested on the MacBook itself. One-time setup on the Mac: Xcode command line tools, Rust, Node.js, and the model download.

Build plan

Four phases, each ending in something that can be shown. Phase 1 is the demo-ready milestone; the rest make it stronger.

Phase	What gets built	Done when	Rough effort
0. Prove the model	Install llama.cpp on the Mac, download the model, chat in the terminal with Wi-Fi off	A good answer appears offline and the speed is acceptable	1 evening
1. Demo app	Tauri app, bundled engine, branded chat screen, presets, offline badge, status bar	The five-minute demo runs start to finish with Wi-Fi off	3–5 days
2. Documents	Drag-and-drop PDF/Word/text, local embeddings, answers with page citations, optional saved history	Questions about a 20-page sample PDF get correct, cited answers	1–2 weeks

Phase	What gets built	Done when	Rough effort
3. Polish and packaging	Model picker, first-run setup that downloads the model, signed and notarized .app, demo script in demo/	A fresh Mac installs and opens it without security warnings	3–5 days

Effort is a rough guess for part-time work and will move once Phase 0 shows real speed on this laptop.

Phase 1 checklist:

- [ ] Create the private repo and scaffold the Tauri app
- [ ] Bundle `llama-server` and start it on 127.0.0.1 when the app opens
- [ ] Chat screen with streamed replies and Stop button
- [ ] Four preset prompts with a system prompt tuned for professional office writing
- [ ] Offline badge and status bar (model, tokens per second)
- [ ] Brand pass using the website's colors and logo
- [ ] Run the demo with Wi-Fi off and fix what breaks

Risks and open questions

Risks:

- **Too slow or too dumb on 16 GB.** Phase 0 tests this in one evening before any app work. Fallback: a smaller model for the live demo, and a recorded clip of the Mac Studio for the "real" tier.
- **Over-promising.** A laptop 8B model is weaker than ChatGPT or Claude. Frame the demo as "private and good enough for everyday office work", and show the bigger box for heavier work.
- **Model licenses.** Stick to Apache 2.0 models (Qwen3, gpt-oss) and keep each model's license file in the repo.
- **macOS security prompts.** An unsigned app shows a warning on other Macs. Fine on Zeke's own laptop; signing needs an Apple Developer account ($99 a year) before sharing it.

Open questions for Zeke:

- [ ] How much RAM does the MacBook Air have (Apple menu › About This Mac)? This picks the model.
- [ ] OK to create a new private repo named `lone-star-private-ai-app`?
- [ ] App name: "Lone Star ITS Private AI" as on the website, or something else?
- [ ] Lead the demo with law and accounting firm examples, per the business plan's suggested first vertical?
- [ ] Will the app ever go to other people's Macs (then sign it), or only run on this laptop for demos?